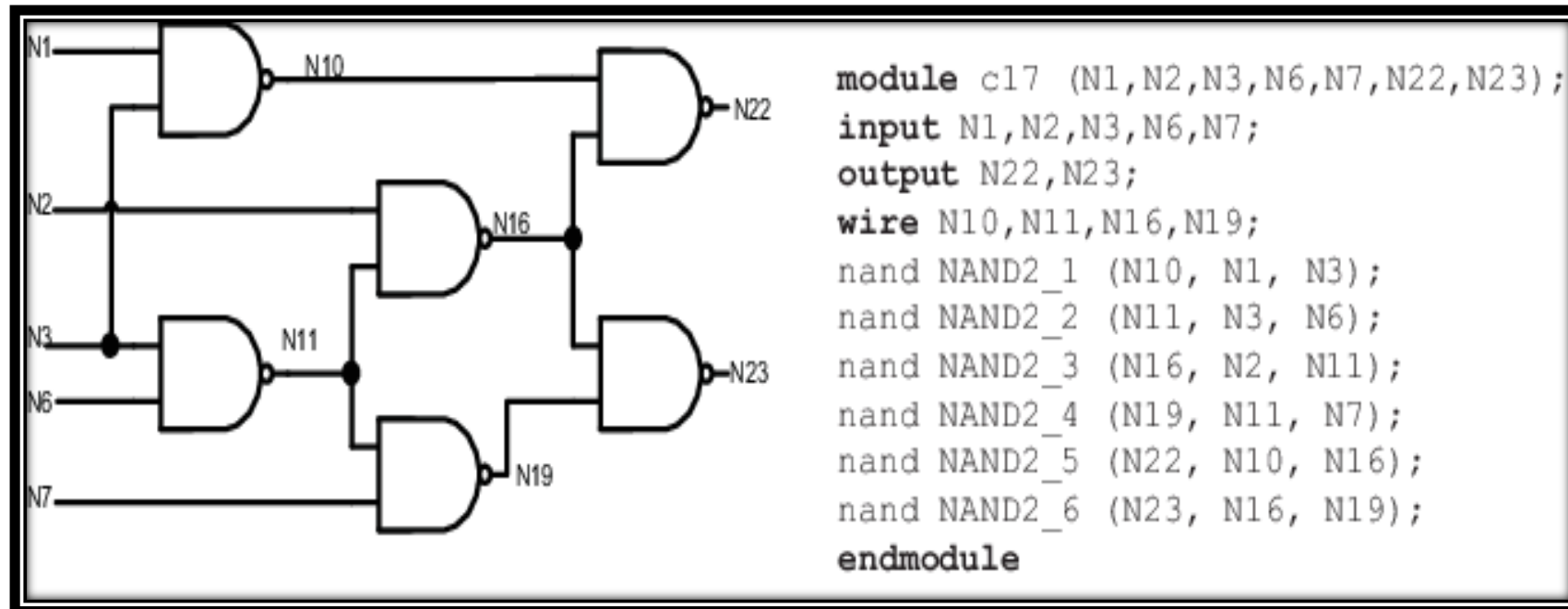




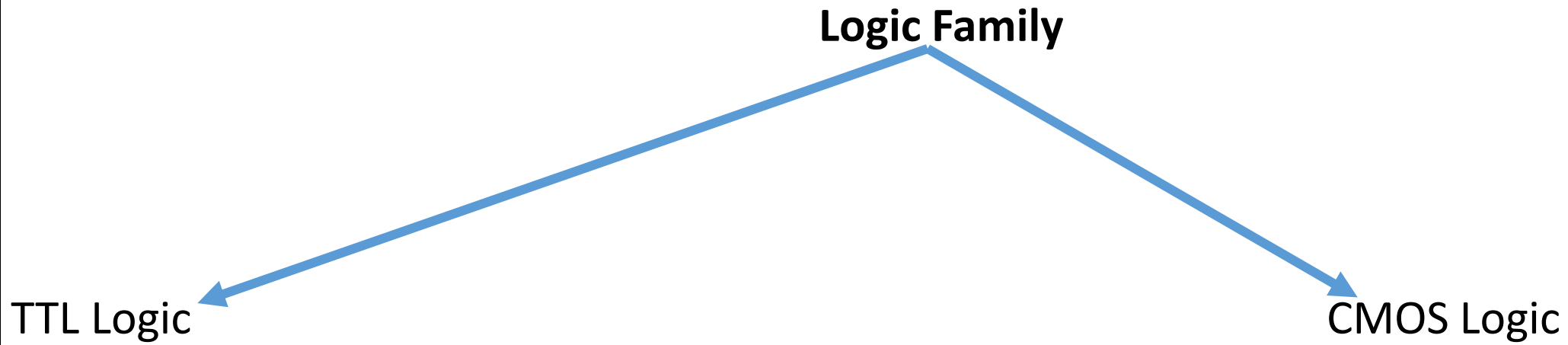
Digital Systems Design : BECE102L

Lecture

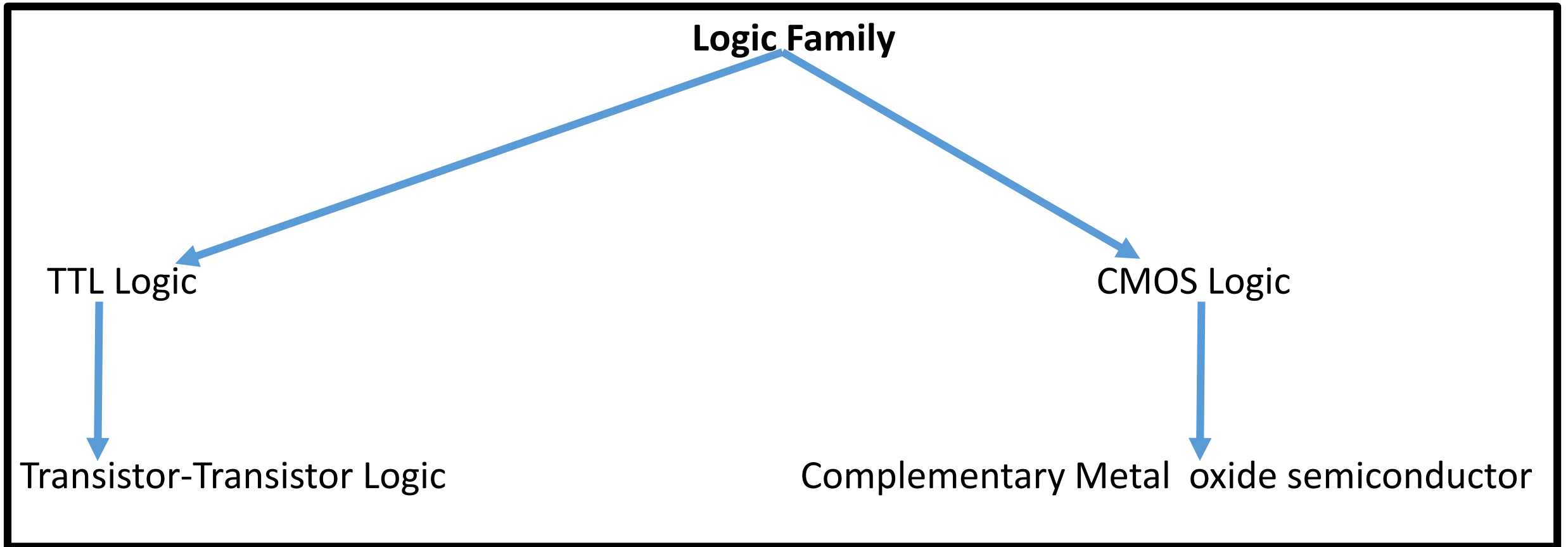
Logic Family



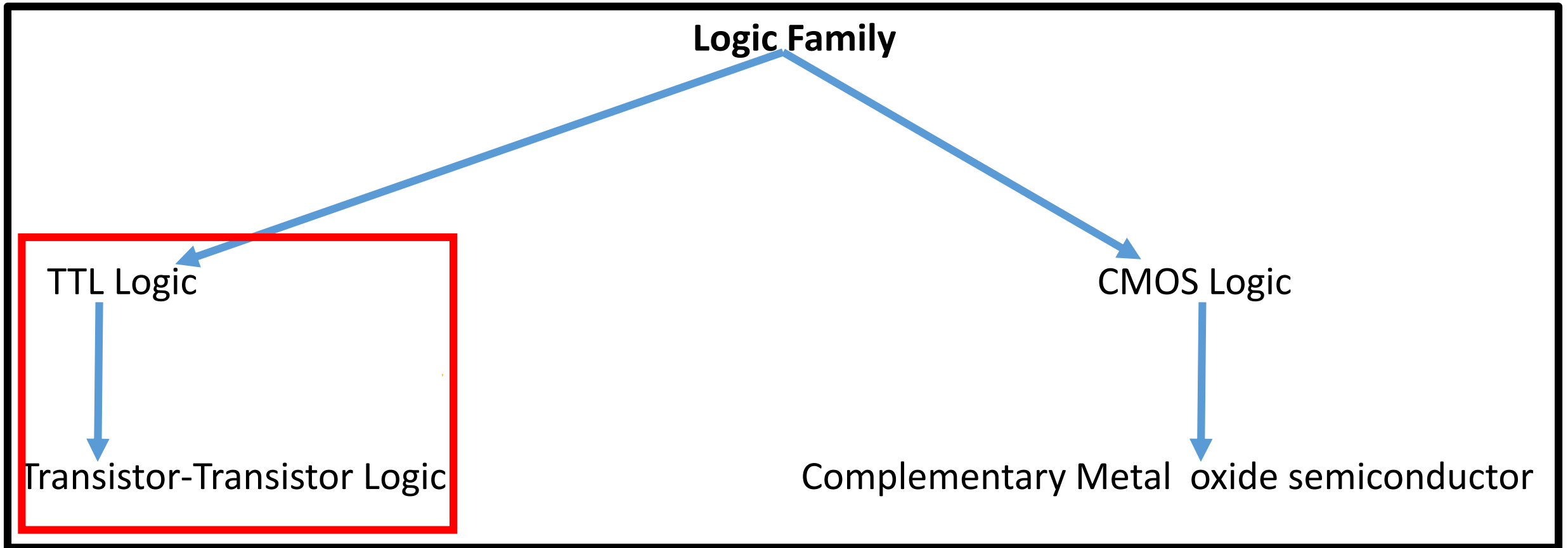
Logic Family



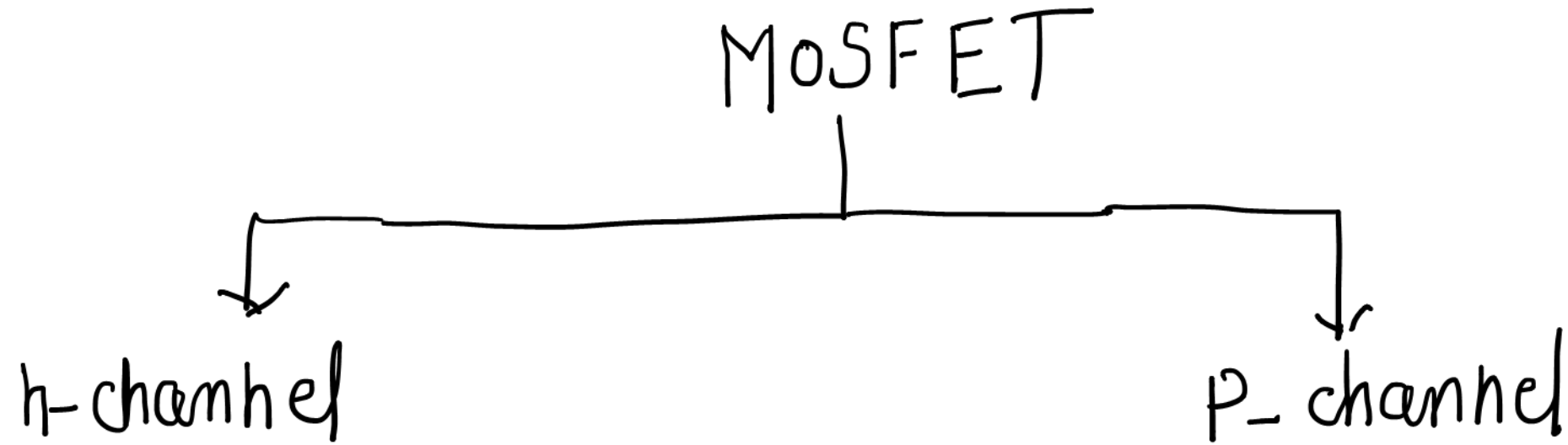
Logic Family



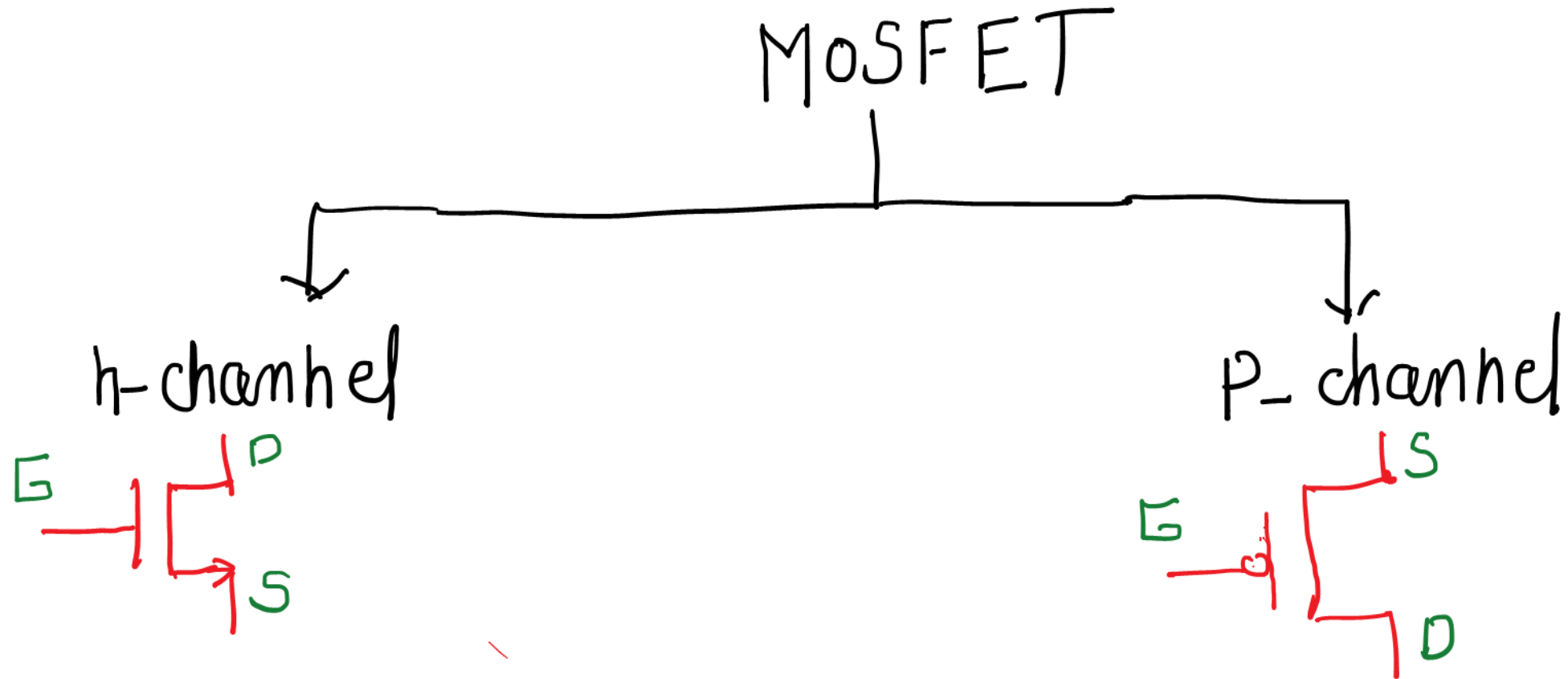
Logic Family



CMOS Logic



CMOS Logic

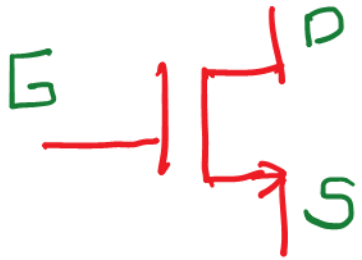


G = gate
S = source

D = Drain

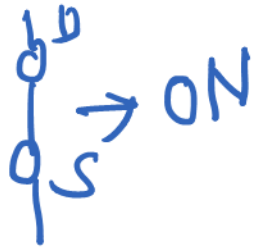
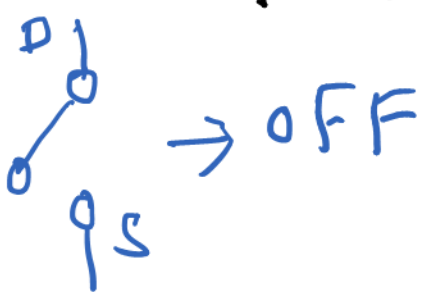
CMOS Logic

n-channel

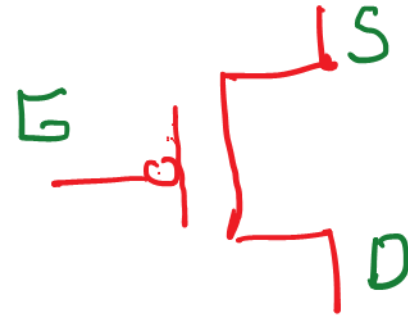


$G = 0 \rightarrow$ transistor OFF

$G = 1 \rightarrow$ transistor



p-channel

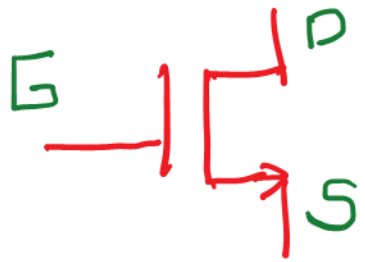


$G =$ gate
 $S =$ source

$D =$ Drain

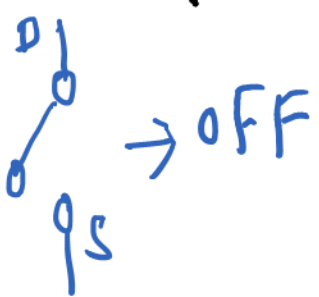
CMOS Logic

n-channel

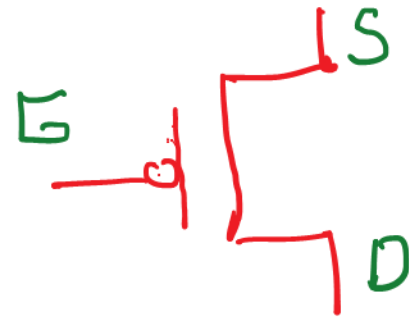


$G = 0 \rightarrow$ transistor off

$G = 1 \rightarrow$ transistor **ON**

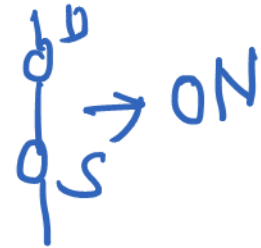
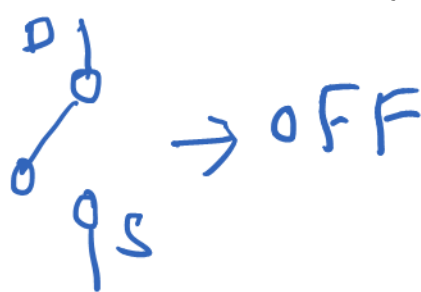


p-channel



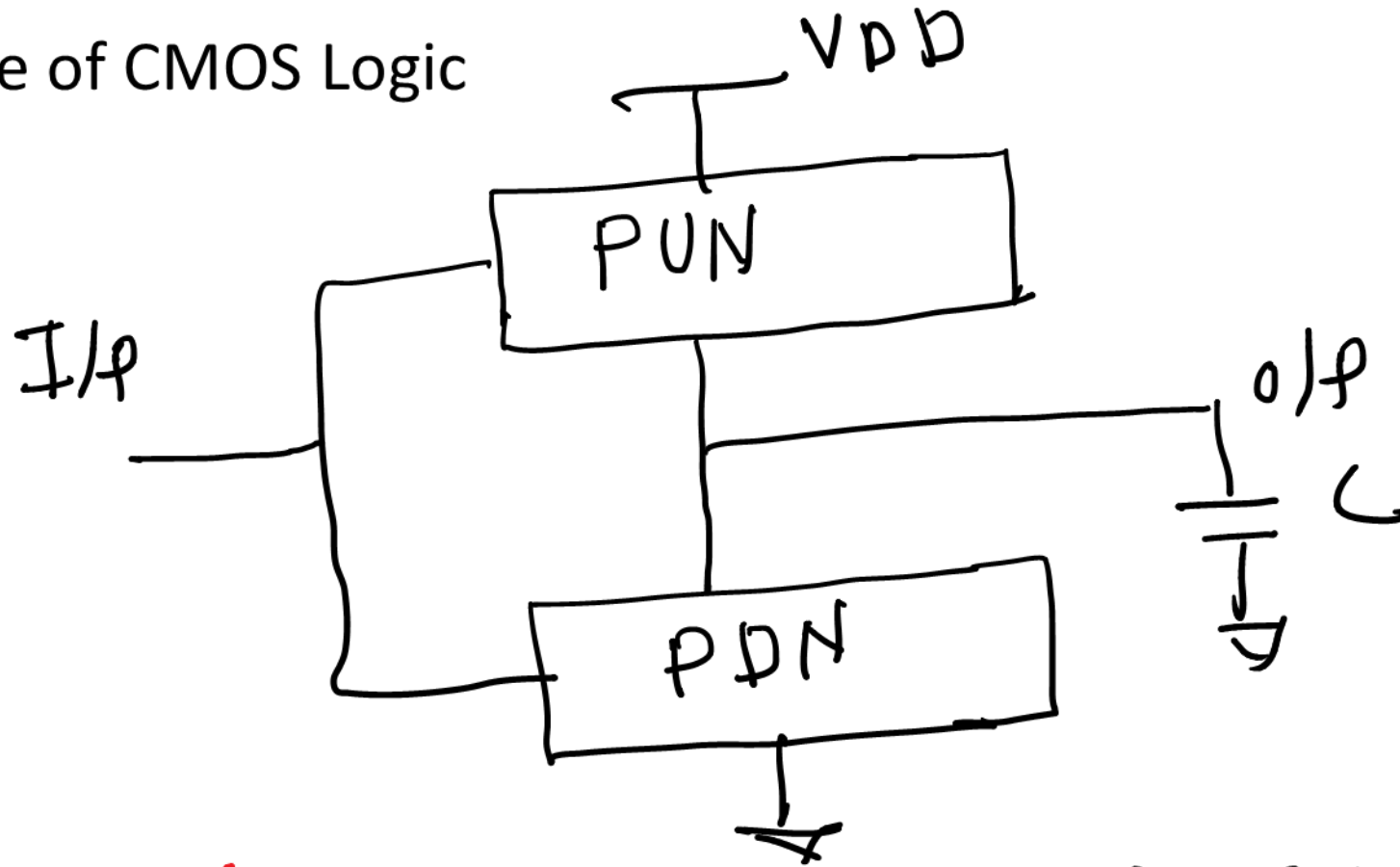
$G = 1 \rightarrow$ transistor off

$G = 0 \rightarrow$ transistor **ON**



CMOS Logic

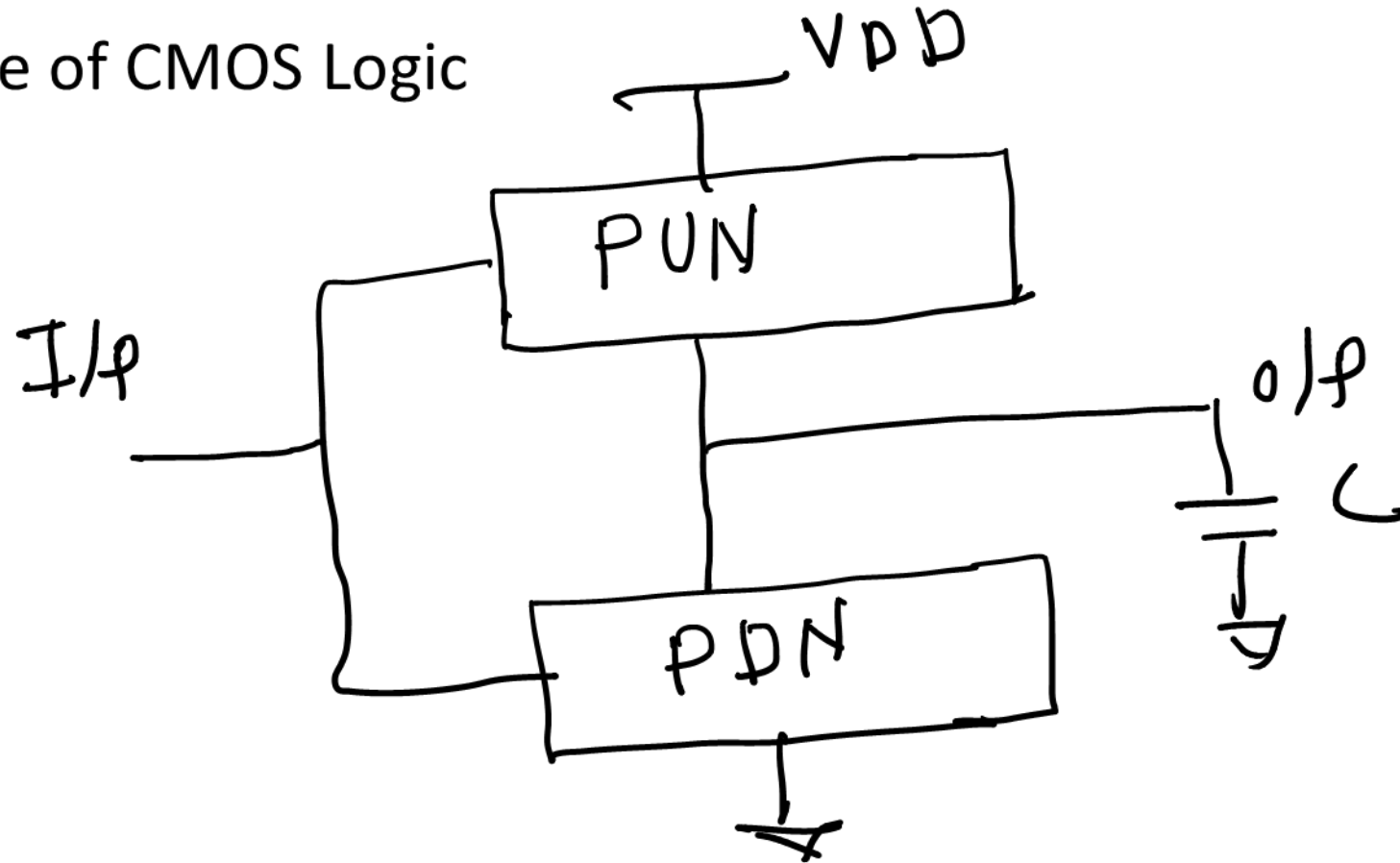
Basic Frame of CMOS Logic



PUN → Pull up Network → use PMOS Transistor

CMOS Logic

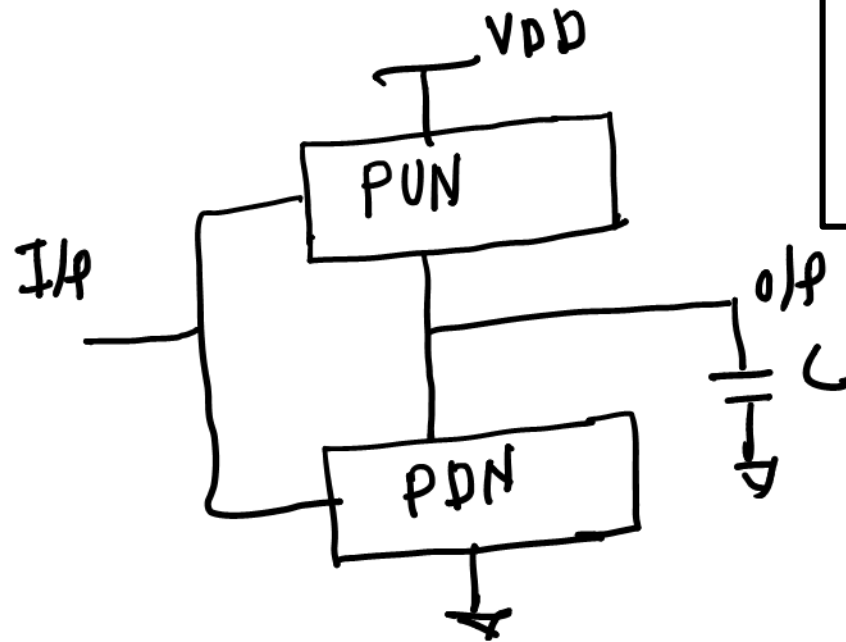
Basic Frame of CMOS Logic



PDN $\rightarrow$ Pull Down network $\rightarrow$ use NMOS N/w

CMOS Logic

Basic Frame of CMOS Logic



PDN → pull Down network. → use NMOS N/w

PDN Rules : NMOS Network

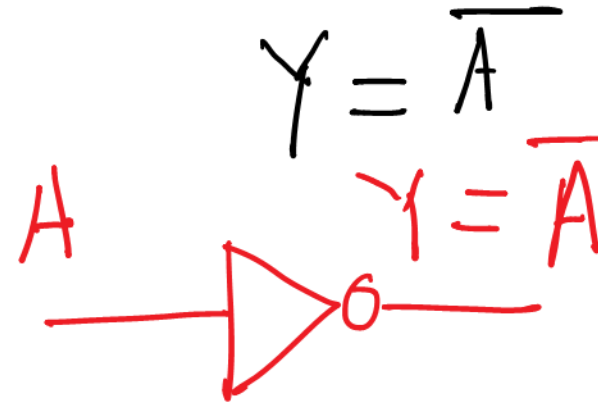
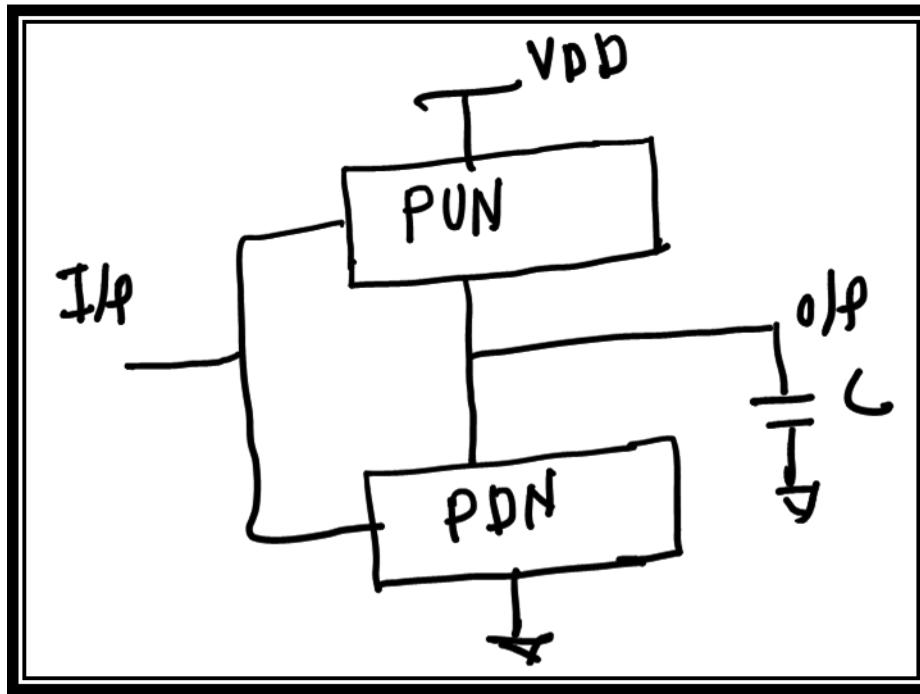
1. (*) Dot product between variables, keep the NMOS transistor in Series
2. (+) Or logic between variables, Keep the PMOS transistor in Parallel

PUN Rules : PMOS Network

1. (+) or between variables, keep the PMOS transistor in Series
2. (*) Dot Product between variables, Keep the PMOS transistor in Parallel

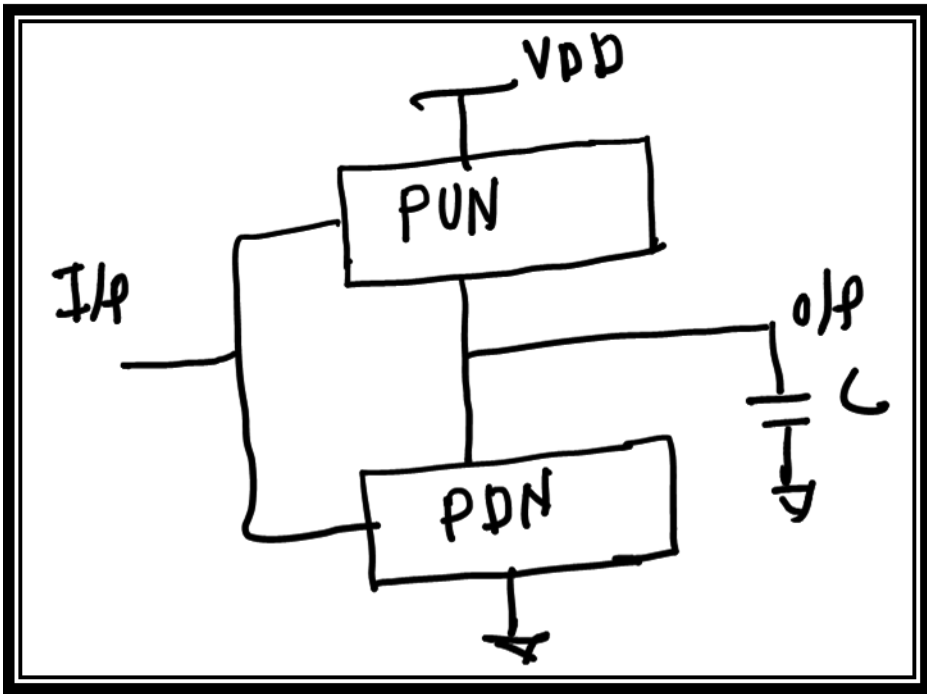
CMOS Logic

CMOS Inverter



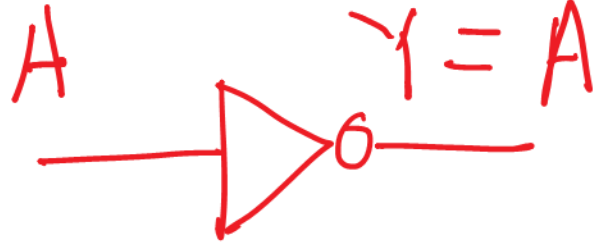
A	Y
0	1
1	0

CMOS Logic



CMOS Inverter

$$Y = \overline{A}$$

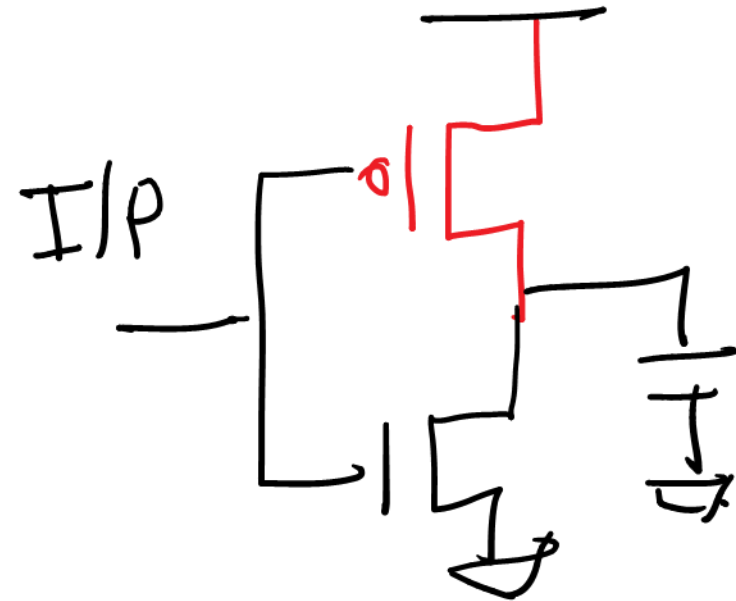
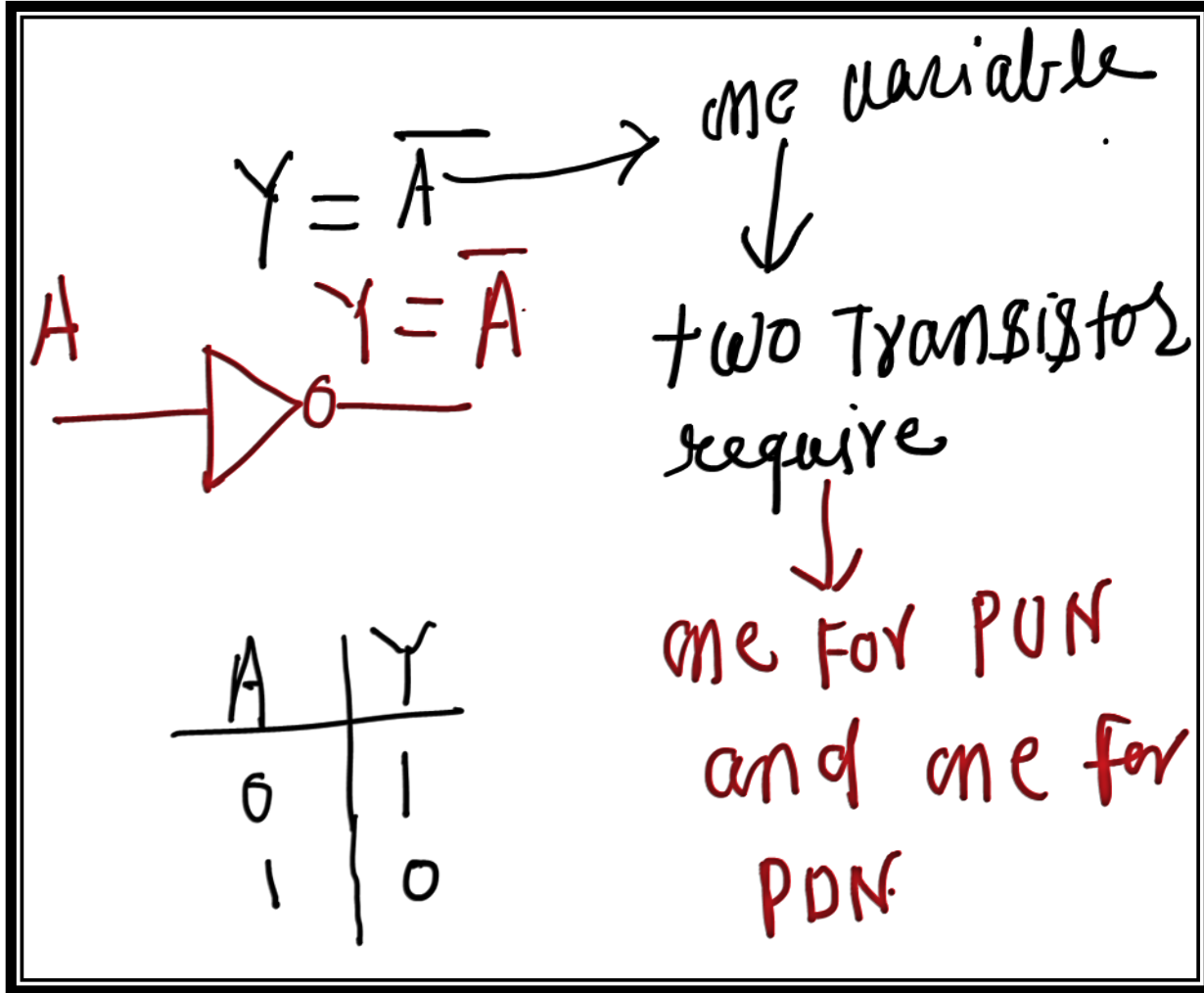


A	Y
0	1
1	0

one variable
↓
two Transistors
require
↓
one for PUN
and one for
PDN.

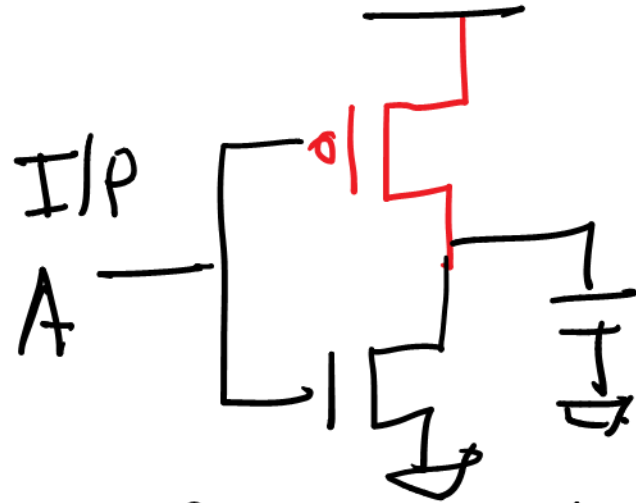
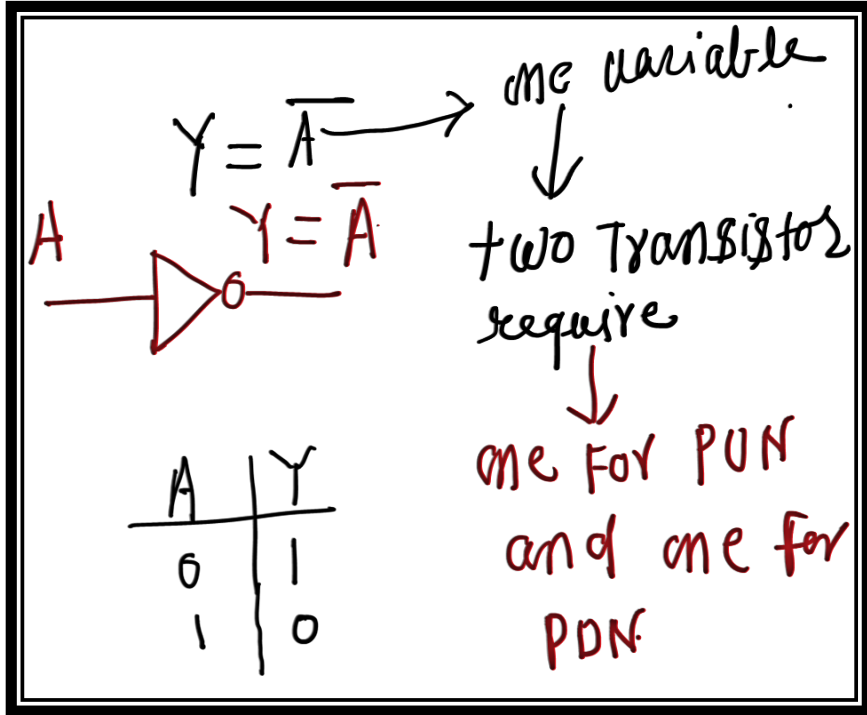
CMOS Logic

CMOS Inverter

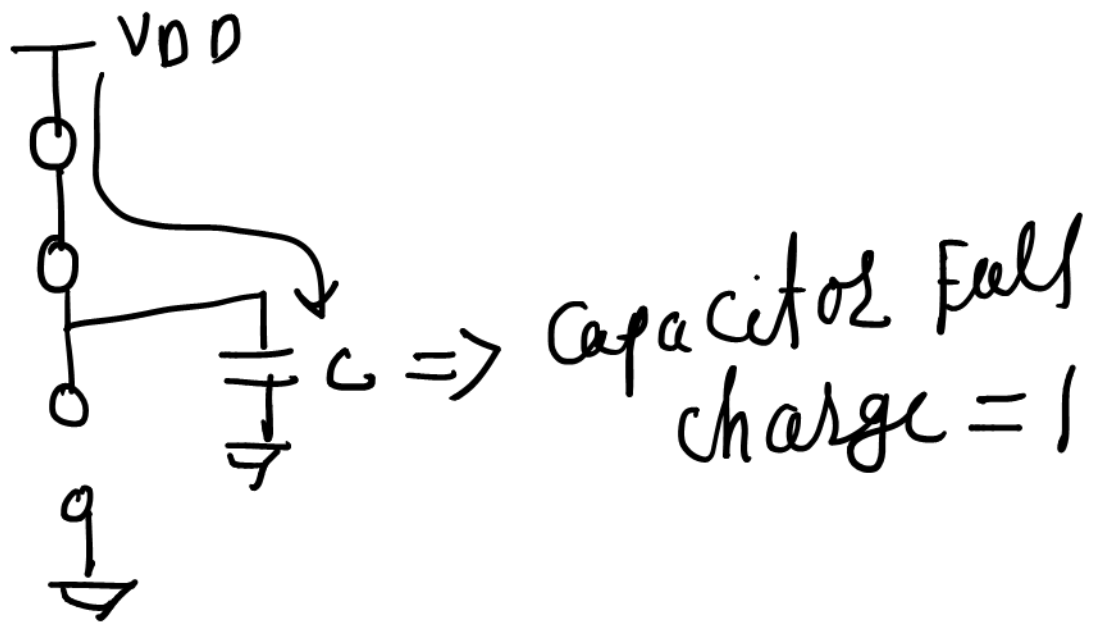


CMOS Logic

CMOS Inverter

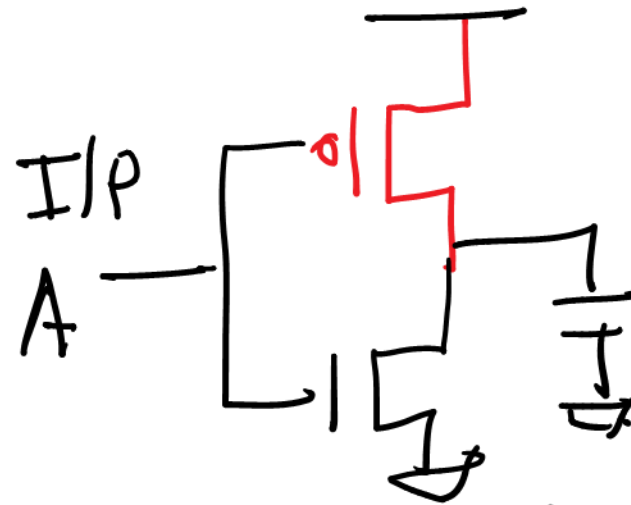
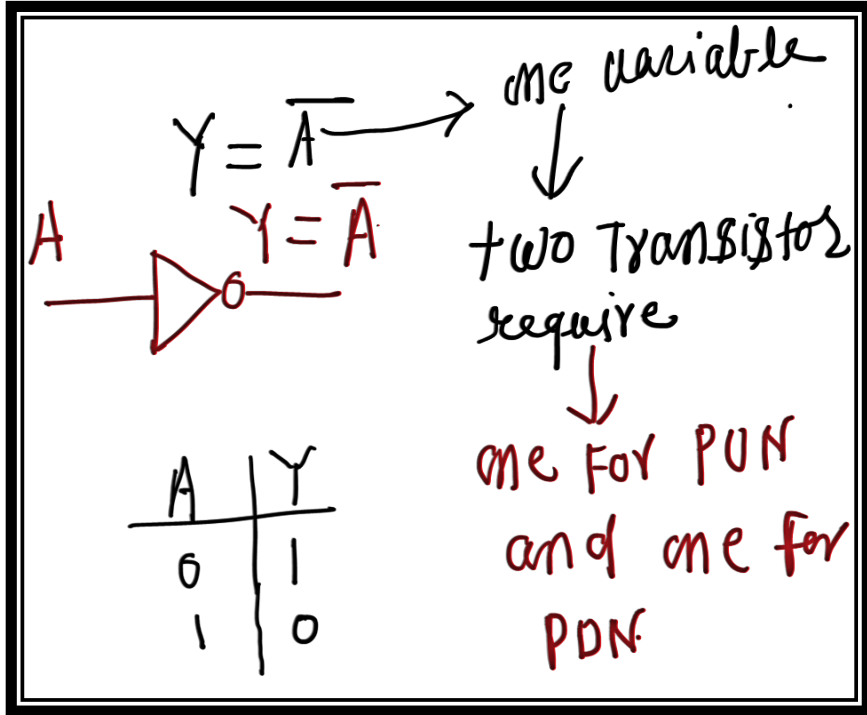


Case 1 :- $A = 0 \Rightarrow I/P = 0$



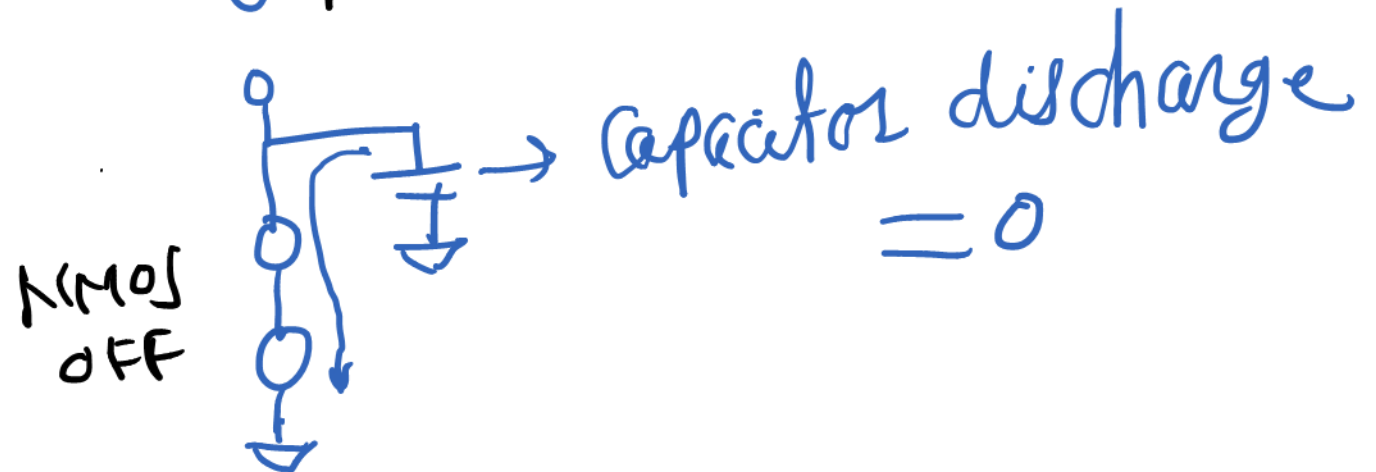
CMOS Logic

CMOS Inverter



Case 2: $A = 1 \Rightarrow I/P = 1$

pmos off



CMOS Logic

NAND Gate

$$Y = \overline{A \cdot B}$$

CMOS Logic

NAND Gate

$$Y = \overline{A \cdot B}$$

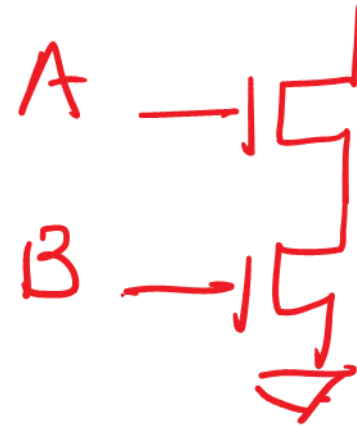
- A & B two variable
- There are 4 Transistors require
- 2 NMOS + 2 PMOS
- (*) dot product mean NMOS network in series

CMOS Logic

NAND Gate

$$Y = \overline{A \cdot B}$$

- A & B two variable
- There are 4 Transistors require
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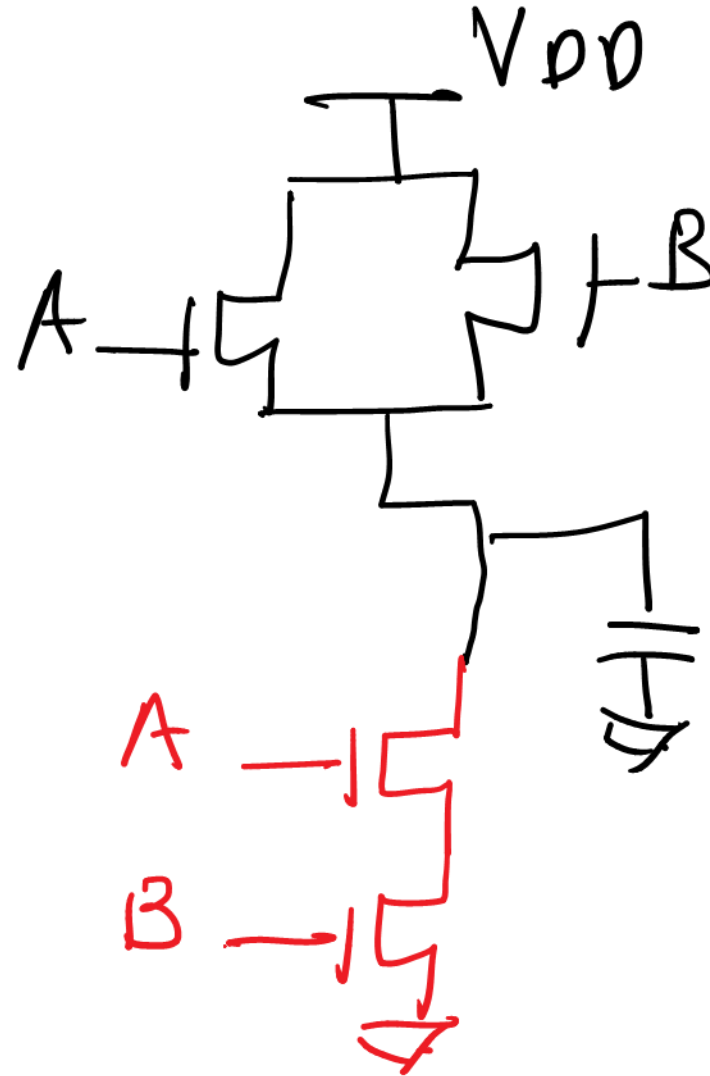


CMOS Logic

NAND Gate

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- There are 4 Transistors require
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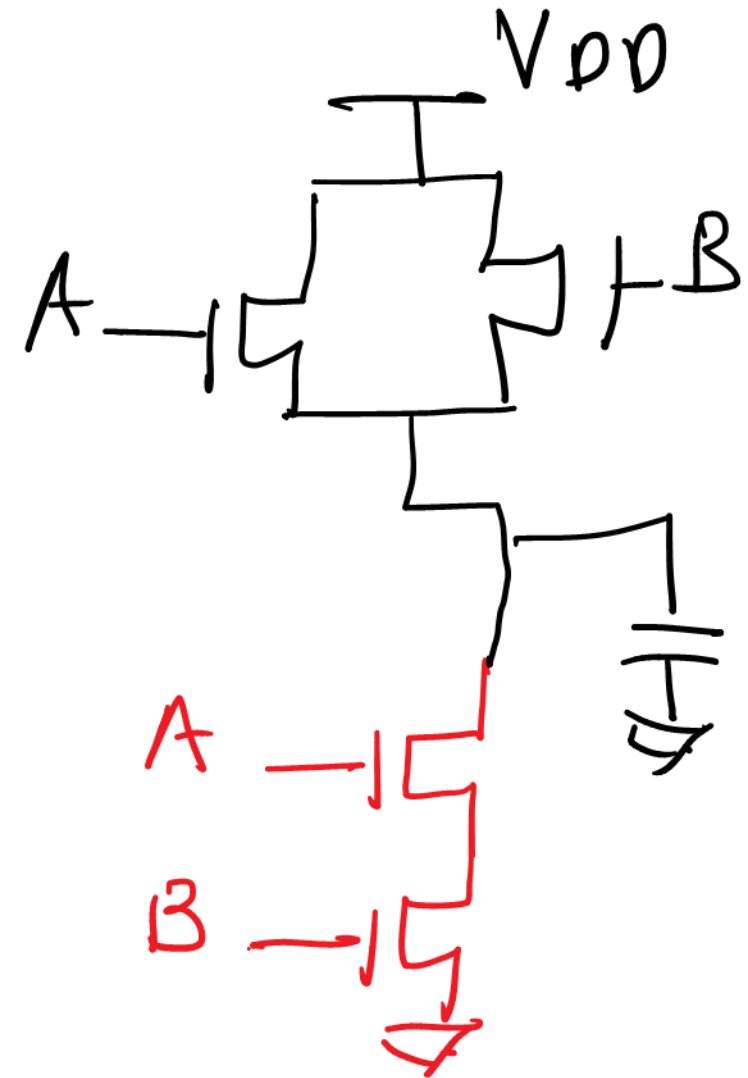


CMOS Logic

NAND Gate

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- There are 4 Transistors require
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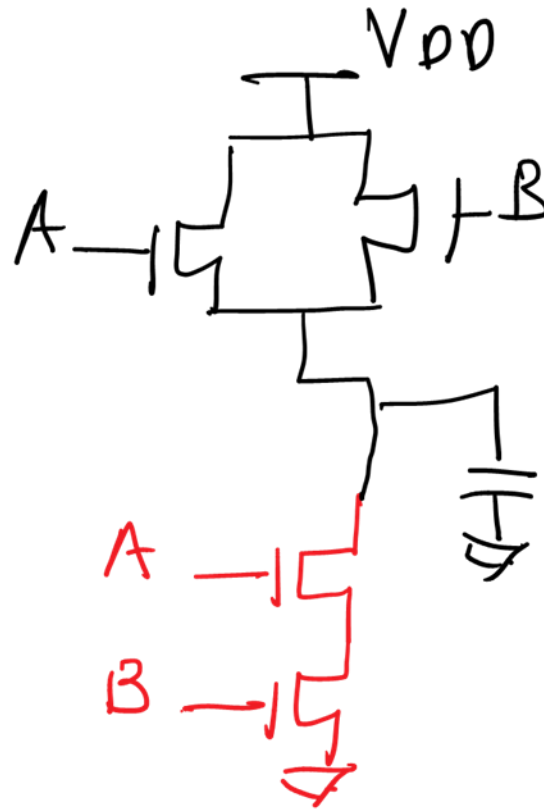


CMOS Logic

NAND Gate

$$Y = \overline{A \cdot B}$$

- A & B two variable
- There are 4 Transistors require
- 2 NMOS + 2 PMOS
- (*) dot product mean NMOS network in series



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

CMOS Logic

NOR Gate

$$Y = \overline{A + B}$$

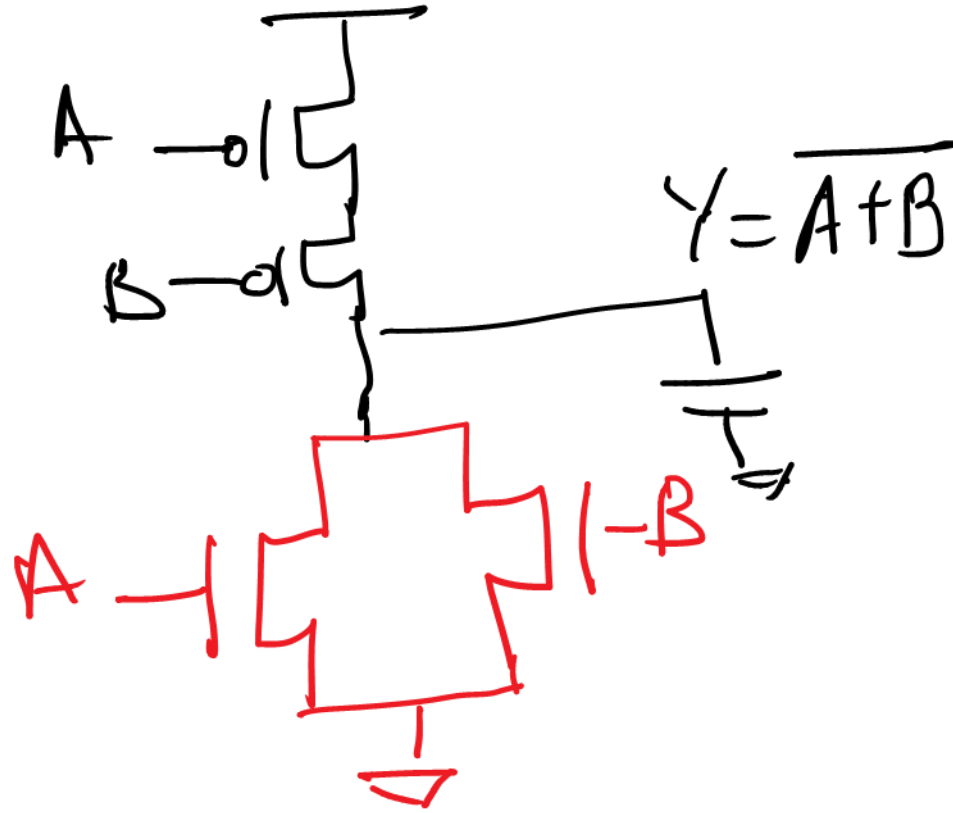
- OR operator
- NMOS network in parallel

CMOS Logic

NOR Gate

$$Y = \overline{A + B}$$

- OR operator
- NMOS Network in parallel

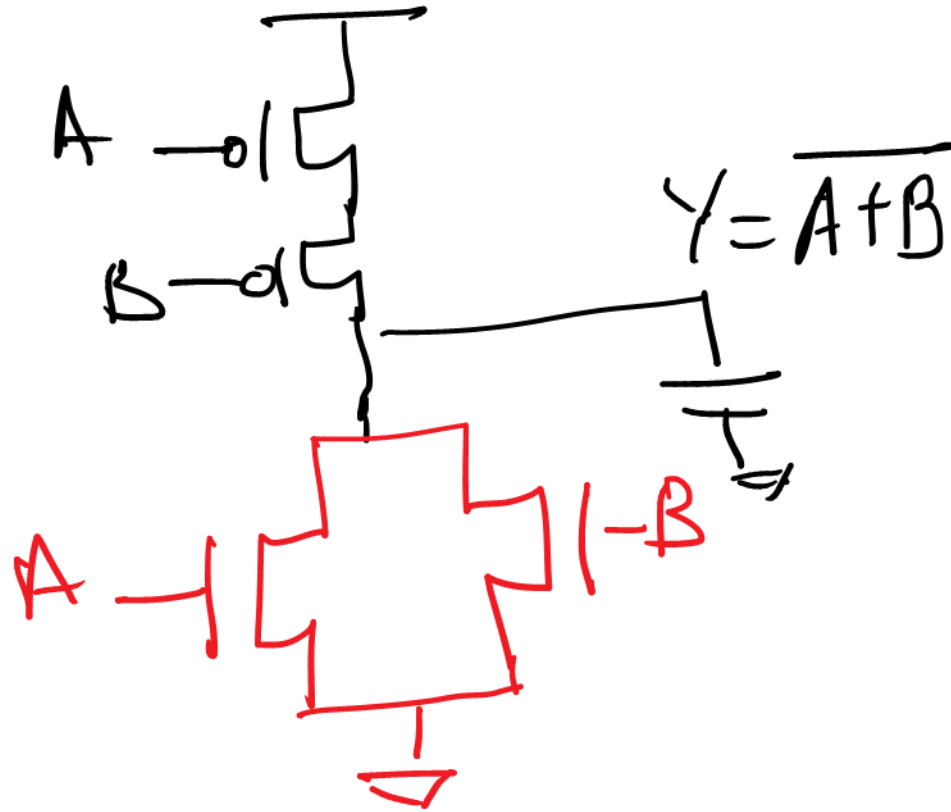
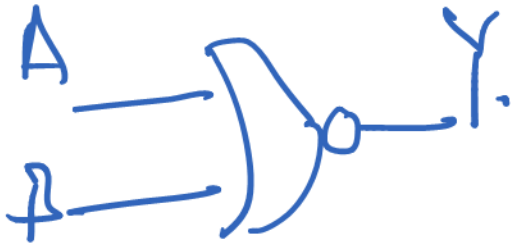


CMOS Logic

NOR Gate

$$Y = \overline{A + B}$$

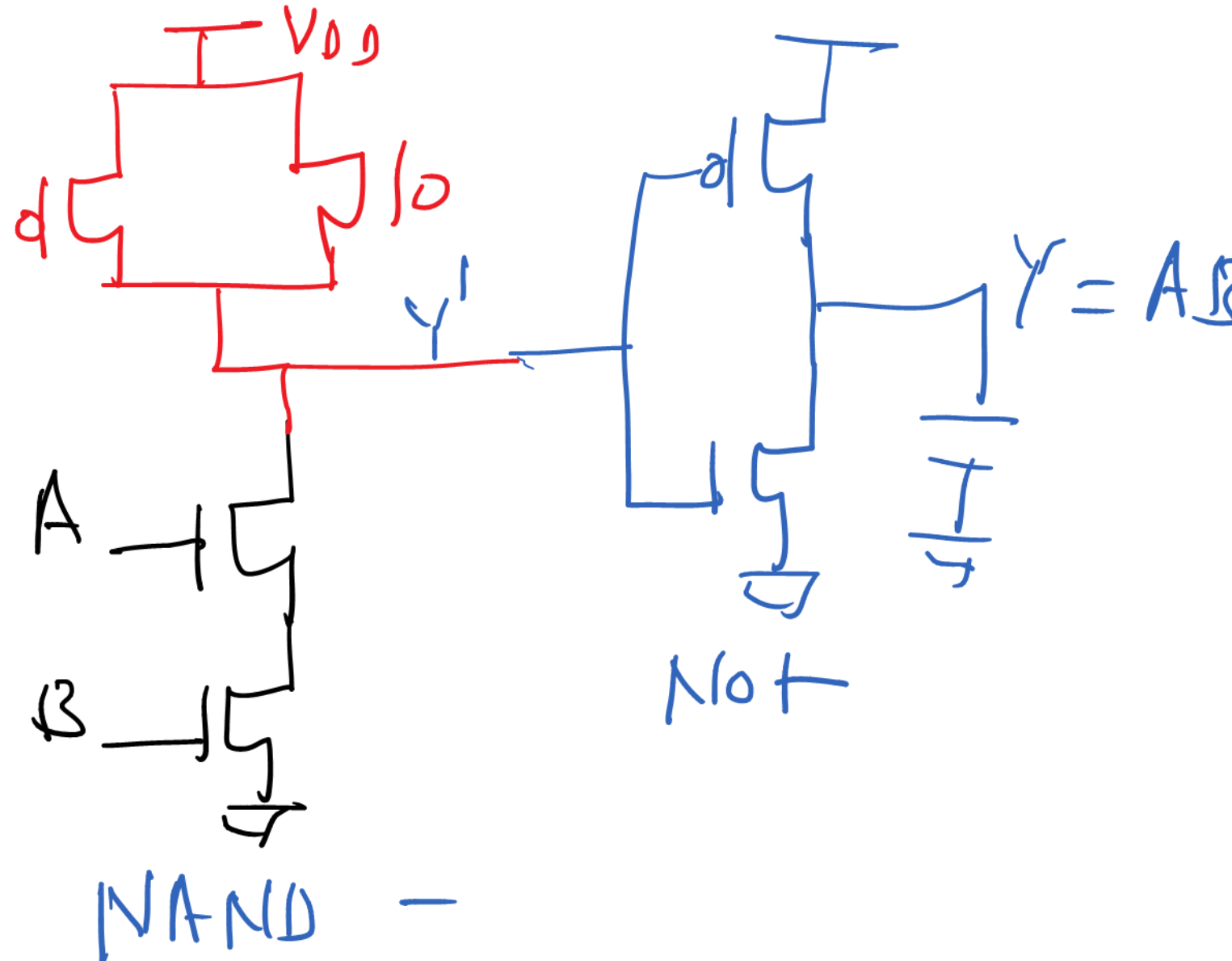
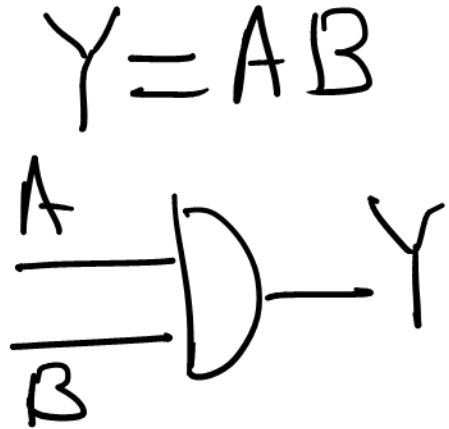
- OR operator
- NMOS Network in parallel



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

CMOS Logic

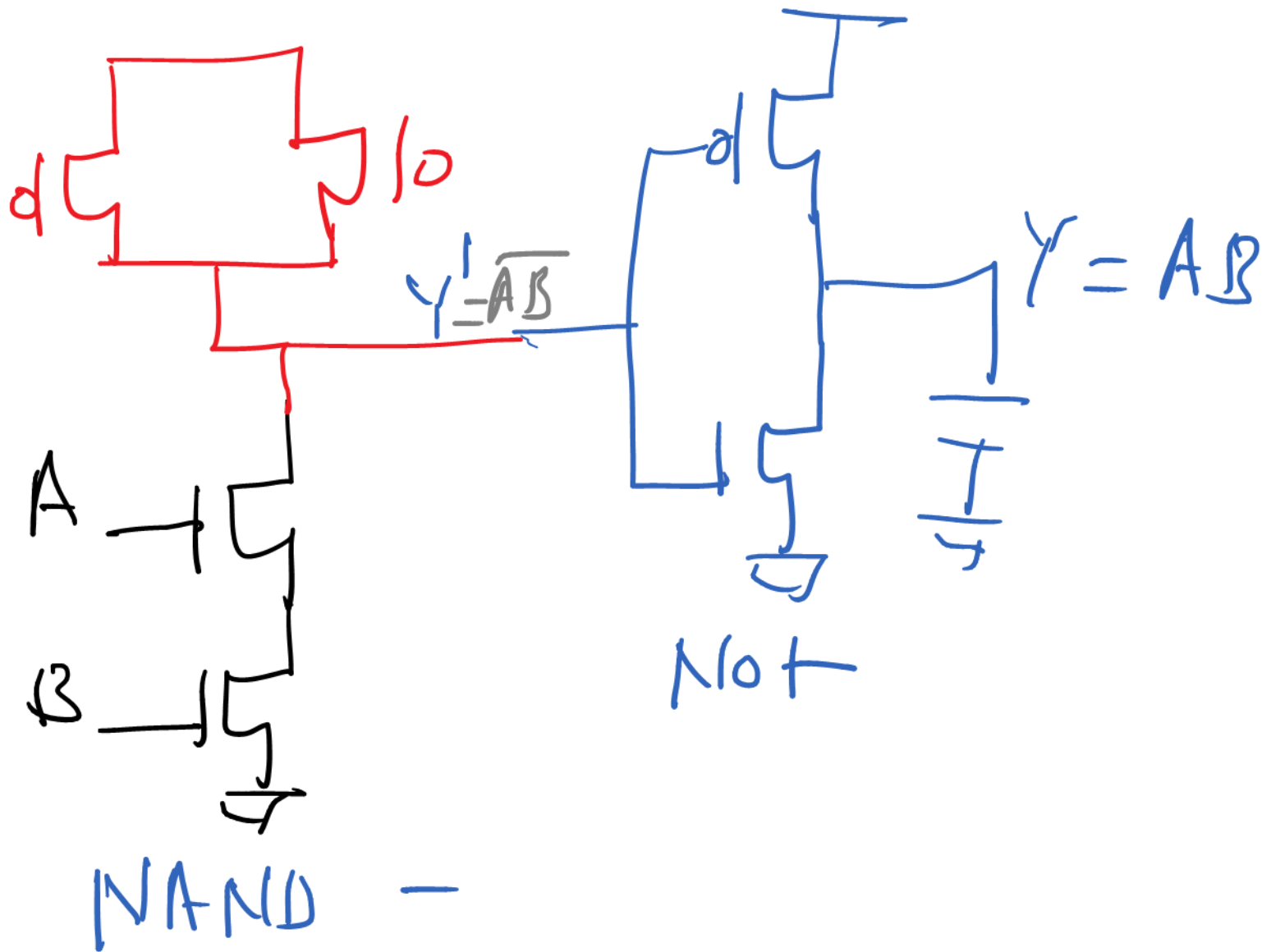
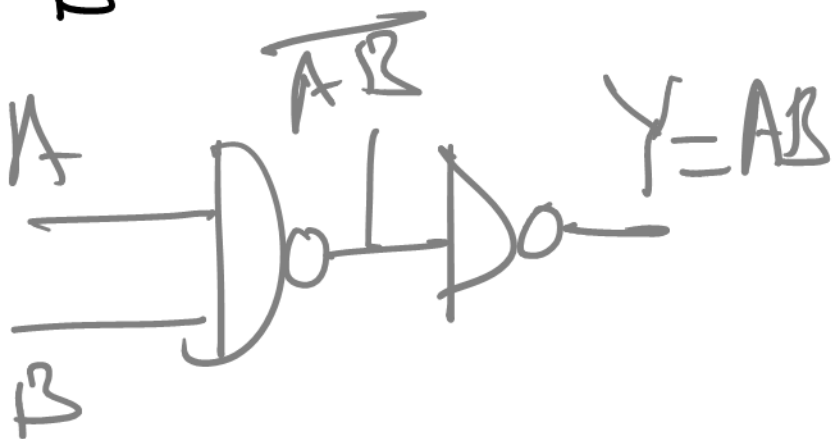
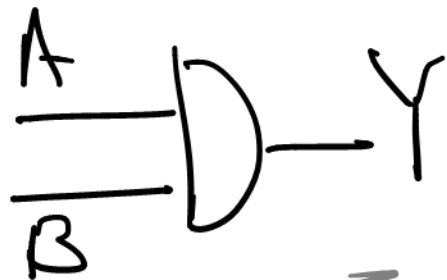
AND Gate



CMOS Logic

AND Gate

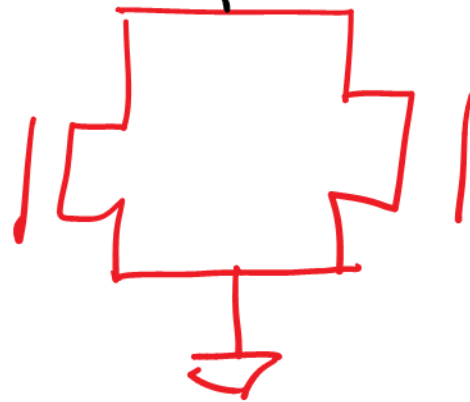
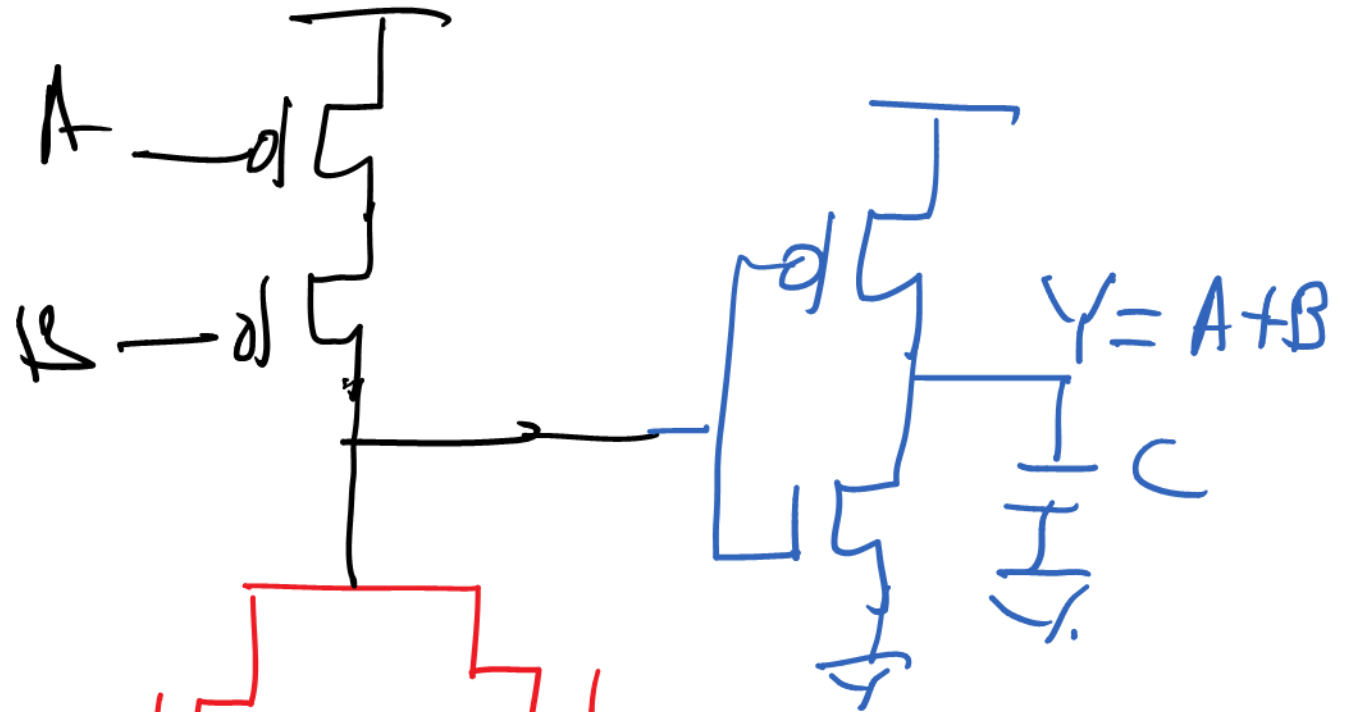
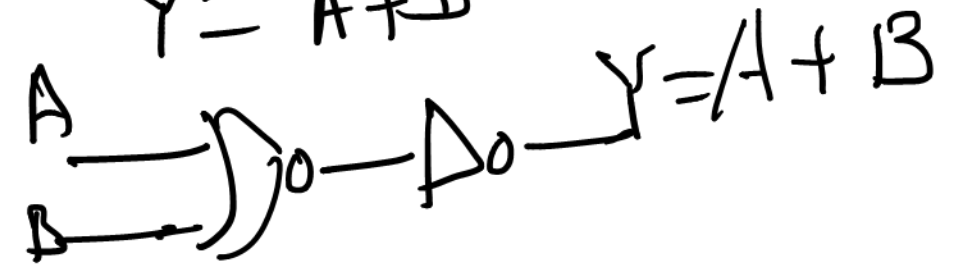
$$Y = AB$$



CMOS Logic

OR gate

$$Y = A + B$$



NOR

Not

CMOS Logic

$$Y = \overline{A + B} C$$

CMOS Logic

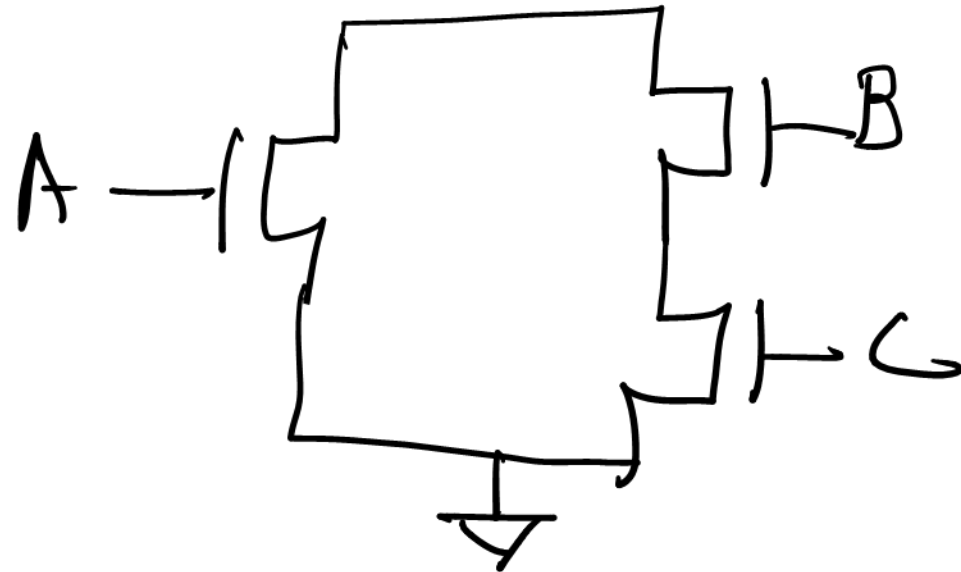
$$Y = \overline{A + BC}$$

↳ 3 variables

6 Transistors req.

↓
3 NMOS

+
3 PMOS



CMOS Logic

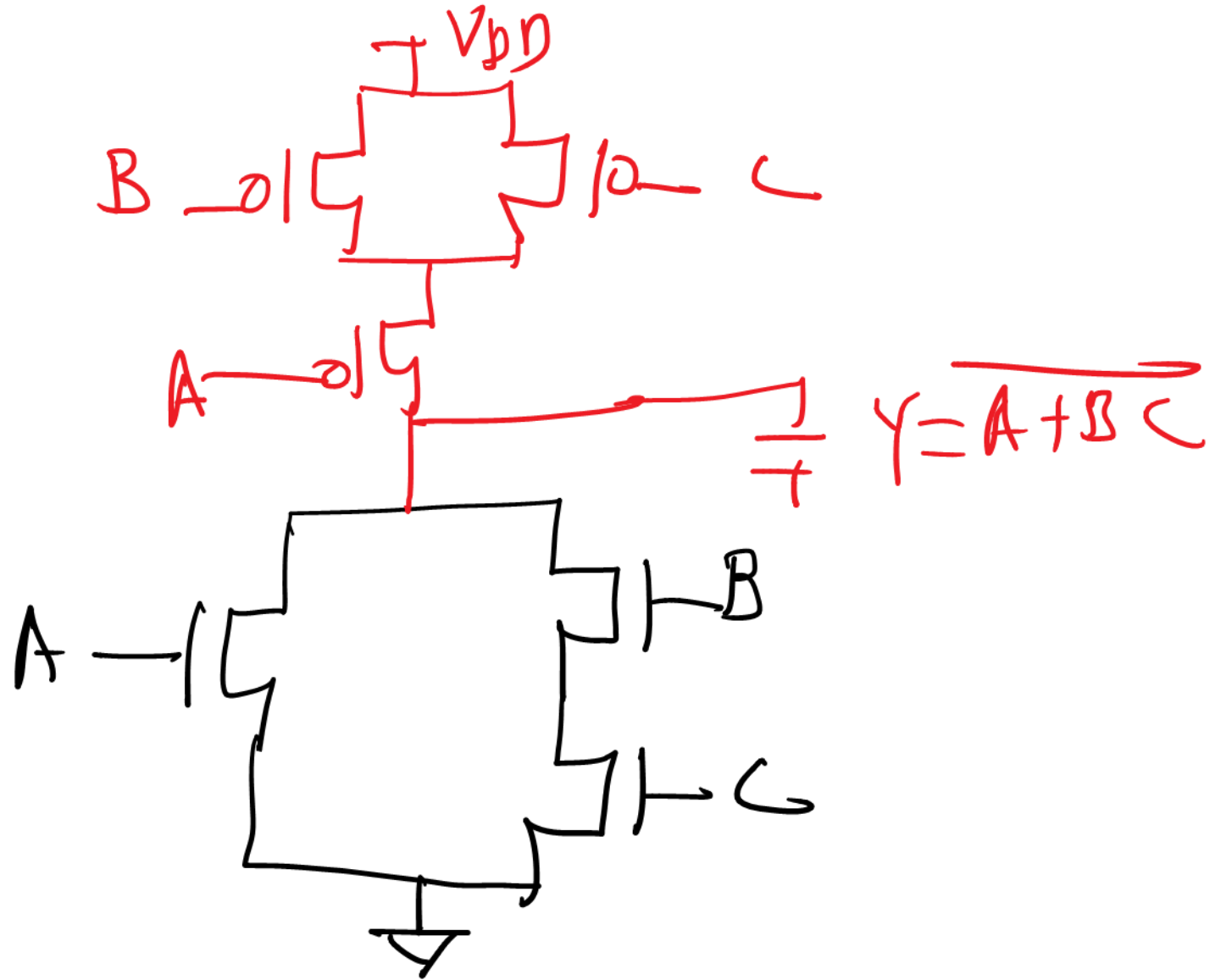
$$Y = \overline{A + B + C}$$

3 variables

6 Transistors req.

3 NMOS

+
3 PMOS



CMOS Logic

NAND Gate

$$Y = \overline{A \cdot B}$$

- A & B two variable
- There are 4 Transistors
require
- 2 NMOS + 2 PMOS

. For two-bit greater-than comparator circuit write

- i. Truth table
- ii. Simplified form
- iii. Gate level Verilog model and verify
- iv. Data flow Verilog model

Hint

First no = $A_0 B_0$

Second No = $A_1 B_1$

objective $\Rightarrow A_0 B_0 > A_1 B_1$

. For two-bit greater-than comparator circuit write

- i. Truth table
- ii. Simplified form
- iii. Gate level Verilog model and verify
- iv. Data flow Verilog model

objective $\Rightarrow A_0 B_0 > A_1 B_1$

[illegible]

. For two-bit greater-than comparator circuit write

- Truth table
- Simplified form
- Gate level Verilog model and verify
- Data flow Verilog model

objective $\Rightarrow A_0 B_0 > A_1 B_1$

Ex. for Y

Ex. 1 $10 > 00 = Y = 1$
(2) (0)

Ex. 2 $10 = 10 = Y = 0$

Ex. 3 $A_0 B_0 < A_1 B_1$
 $00 < 11 \Rightarrow Y = 0$
(0) (3)

A_0	B_0	A_1	B_1	Y
0	0	0	0	
0	0	0	1	
0	0	1	0	
0	0	1	1	
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	

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 $(2) \quad (0)$

Ex. 2 $10 = 10 = Y = 0$

Ex. 3 $A_0 B_0 < A_1 B_1 \Rightarrow Y = 0$
 $(0) \quad (3)$

A_0	B_0	A_1	B_1	Y
0	0	0	0	
0	0	0	1	
0	0	1	0	
0	0	1	1	0
0	1	0	0	
0	1	0	1	
0	1	1	0	
0	1	1	1	
1	0	0	0	1
1	0	0	1	
1	0	1	0	0
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	

In The same
Manner prepare out put of
Truth table and solve using
4 variable K-Map